

GENERAL REQUIREMENTS

A. GENERAL

1. SPECIFICATIONS ARE APPLICABLE TO ALL CONTRACTORS AND/OR SUBCONTRACTORS FOR MECHANICAL SYSTEMS.
2. CHECK OTHER PLANS AND SPECIFICATIONS AND FULLY COORDINATE WITH OTHER TRADES, OWNER AND ARCHITECT REQUIREMENTS.
3. CONFORM TO ALL GENERAL AND SPECIAL CONDITIONS OF CONTRACT AS SPECIFIED BY ARCHITECT AND/OR OWNER.
4. VISIT SITE, CHECK FACILITIES AND CONDITIONS AND MAKE ALL NECESSARY OBSERVATIONS AND MEASUREMENTS. NOTE CONDITIONS UNDER WHICH WORK IS TO BE PERFORMED AND TAKE ALL ITEMS INTO CONSIDERATION IN BID.
5. SYSTEMS ARE TO BE COMPLETE AND WORKABLE IN ALL RESPECTS, PLACED IN OPERATION AND PROPERLY ADJUSTED.
6. EACH CONTRACTOR SHALL PROVIDE HIS OWN CLEAN-UP, REMOVAL AND LEGAL DISPOSAL OF ALL RUBBISH DAILY.
7. CONTRACTOR SHALL PROTECT NEW WORK, EXISTING WORK AND ADJACENT PROPERTY AGAINST WEATHER.
8. CONTRACTOR SHALL PROTECT HIS WORK, MATERIALS, APPARATUS AND FIXTURES FROM DAMAGE. ANY WORK DAMAGED BY FAILURE TO PROVIDE PROTECTION REQUIRED, SHALL BE REMOVED AND REPLACED WITH NEW MATERIAL AT THE CONTRACTOR'S EXPENSE.
9. CONTRACTOR MUST CONFIRM ALL UTILITY COMPANY REQUIREMENTS AND CONNECTION POINTS IN FIELD PRIOR TO STARTING WORK.
10. ARRANGE FOR AND OBTAIN OWNER'S AND INSURANCE REPRESENTATIVE'S PERMISSION FOR ANY SERVICE SHUTDOWNS.
11. THE CONTRACTOR SHALL BE SOLELY RESPONSIBLE FOR CONSTRUCTION MEANS, METHODS, SEQUENCES OF CONSTRUCTION AND THE SAFETY OF WORKERS.
12. NO PIPING, DUCTWORK, WIRING, ETC. SHALL BE INSTALLED OR ROUTED ABOVE ELECTRICAL PANELS AND EQUIPMENT.
13. THE MECHANICAL CONTRACTOR SHALL COORDINATE WITH THE ELECTRICAL CONTRACTOR AND OBTAIN A WRITTEN APPROVAL IDENTIFYING THE ELECTRICAL CHARACTERISTICS OF ALL MECHANICAL EQUIPMENT PRIOR TO ORDERING OF EQUIPMENT. NO ADDITIONAL PAYMENT WILL BE MADE FOR LACK OF CONTRACTOR COORDINATION OF ELECTRICAL CHARACTERISTICS.
14. DURING CONSTRUCTION THE CONTRACTOR MAY UNCOVER AN EXISTING CONDITION THAT WILL HAVE TO BE MODIFIED. ANY SUCH WORK WHICH COMES UNDER THE JURISDICTION OF THIS CONTRACTOR SHALL BE DONE BY THIS CONTRACTOR WITHOUT EXTRA COST TO THE OWNER, AS THOUGH FULLY DETAILED ON PLANS AND/OR DESCRIBED IN THE SPECIFICATIONS.

B. CODES, PERMITS, STANDARDS AND REGULATIONS

1. CONFORM TO ALL APPLICABLE CODES (LOCAL, STATE, NATIONAL CODES, NFPA, OSHA, ETC.). GOVERNMENT REGULATIONS, UTILITY COMPANY REQUIREMENTS, AND APPLICABLE STANDARDS.
2. OBTAIN PERMITS AND PAY ALL FEES ARRANGE FOR ALL REQUIRED INSPECTIONS AND APPROVALS.
- C. RELATED WORK SPECIFIED ELSEWHERE
1. OPENINGS AND CHASES, WHEN SHOWN ON ARCHITECTURAL DRAWINGS.
2. TEMPORARY WATER SERVICE, SANITARY FACILITIES, AND HEATING DURING CONSTRUCTION.
3. POURED-IN-PLACE CONCRETE.
4. FINISHED PAINTING.
5. ELECTRIC POWER WIRING
- D. DRAWINGS
1. THE SYSTEMS AS SHOWN ON MECHANICAL DRAWINGS ARE DIAGRAMMATIC. CONFIRM ALL DIMENSIONS BY FIELD MEASUREMENT.
2. THE EXACT LOCATIONS FOR FIXTURES, EQUIPMENT AND PIPING WHICH IS NOT COVERED BY DRAWINGS, SHALL BE OBTAINED FROM THE ARCHITECT OR HIS REPRESENTATIVE IN THE FIELD AND THE WORK SHALL BE LAID OUT ACCORDINGLY.
3. DRAWINGS AND SPECIFICATIONS ARE INTENDED TO SUPPLEMENT ONE ANOTHER. ANY MATERIALS OR LABOR CALLED FOR IN ONE BUT NOT THE OTHER SHALL BE FURNISHED AS IF BOTH WERE MENTIONED IN THE SPECIFICATIONS AND SHOWN ON THE DRAWINGS.

E. BASE EQUIPMENT, MATERIALS AND SUBSTATIONS.

1. ALL EQUIPMENT AND MATERIALS SHALL BE NEW, FREE OF DEFECTS AND UL LABELED.
2. BASE BID MANUFACTURERS ARE INCLUDED IN THE SPECIFICATIONS OR LISTED IN SCHEDULES ON THE DRAWINGS. ALL OTHER MANUFACTURERS ARE CONSIDERED A SUBSTITUTION.
3. THE NAME OR MAKE OF ANY ARTICLE, DEVICE, MATERIAL, FORM OF CONSTRUCTION, FIXTURE, ETC. STATED IN THIS SPECIFICATIONS, WHETHER OR NOT THE WORDS "OR APPROVED EQUAL" ARE USED, SHALL BE KNOWN AS A "STANDARD".
4. ALL PROPOSALS SHALL BE BASED ON "STANDARDS" SPECIFIED.
5. THE EQUIPMENT SCHEDULES ON THE DRAWINGS INDICATE MANUFACTURERS EQUIPMENT MODEL NUMBERS THAT THIS DESIGN HAS BEEN BASED ON. THE USE OF OTHER MANUFACTURERS' EQUIPMENT THAT IS LISTED AS ACCEPTABLE ALTERNATES THAT ENTAILS GENERAL TRADES, STRUCTURAL, MECHANICAL, ELECTRICAL, ETC. REVISIONS IS THIS CONTRACTOR'S RESPONSIBILITY. ANY ADDITIONAL COST OF SUCH CHANGES SHALL BE PAID BY THE CONTRACTOR SUBMITTING THE ACCEPTABLE ALTERNATES WHICH NECESSITATES CHANGES IN INSTALLING SUCH SUBMITTED ALTERNATE EQUIPMENT, EVEN THOUGH SUCH COSTS MAY BE PART OF ANOTHER DIVISION OF WORK.
6. SUBSTITUTIONS ARE SUBJECT TO THE APPROVAL OF THE OWNER. IF A SUBSTITUTION IS SUBMITTED, IT IS THE CONTRACTOR'S RESPONSIBILITY TO EVALUATE IT AND CERTIFY THAT THE SUBSTITUTION IS EQUIVALENT IN ALL RESPECTS TO THE BASE SPECIFICATIONS.
7. IF SUBSTITUTIONS ARE APPROVED, NOTIFY ALL OTHER CONTRACTORS, SUBCONTRACTORS, ETC. AFFECTED BY THE SUBSTITUTION AND FULLY COORDINATE WITH THEM. ANY COSTS RESULTING FROM SUBSTITUTION, WHETHER BY THIS CONTRACTOR OR OTHERS, SHALL BE THE RESPONSIBILITY OF AND PAID FOR BY THE SUBSTITUTING CONTRACTOR. APPROVED SHOP DRAWINGS DO NOT ABSOLVE THIS CONTRACTOR FROM THIS RESPONSIBILITY.
8. ALL EQUIPMENT SHALL BE INSTALLED IN FULL ACCORDANCE WITH THE MANUFACTURER'S DATA AND INSTALLATION INSTRUCTIONS. IT IS THIS CONTRACTOR'S RESPONSIBILITY TO CHECK AND CONFIRM THESE REQUIREMENTS PRIOR TO STARTING WORK.

F. WARRANTY

1. FULLY WARRANT ALL MATERIALS, EQUIPMENT AND WORKMANSHIP FOR (1) YEAR FROM DATE OF ACCEPTANCE.
2. EXTEND ALL MANUFACTURERS' WARRANTIES TO OWNER, INCLUDING FIVE YEAR (5) COMPRESSOR AND TEN (10) YEAR HEAT EXCHANGER EXTENDED WARRANTY ON HVAC EQUIPMENT.
3. REPAIR OR REPLACE WITHOUT CHARGE TO THE OWNER ALL ITEMS FOUND DEFECTIVE DURING THE WARRANTY PERIOD. IN THE CASE OF REPLACEMENT OR REPAIR DUE TO FAILURE WITHIN THE WARRANTY PERIOD, THE WARRANTY ON THAT PORTION OF THE WORK SHALL BE EXTENDED FOR A MINIMUM PERIOD OF ONE (1) YEAR FROM THE DATE OF SUCH REPLACEMENT OR REPAIR.

G. SHOP DRAWING SUBMITTALS

1. SHOP DRAWINGS FOR MECHANICAL AND PLUMBING SYSTEMS, INCLUDING BUT NOT LIMITED TO SHEET METAL, PLUMBING FIXTURES AND EQUIPMENT WITH ADEQUATE DETAILS AND SCALES TO CLEARLY SHOW CONSTRUCTION. INDICATE THE OPERATING CHARACTERISTICS FOR EACH REQUIRED ITEM. CLEARLY IDENTIFY EACH ITEM ON THE SUBMITTAL AS TO MARK, LOCATION AND USE, USING SAME IDENTIFICATION AS PROVIDED ON THE CONSTRUCTION DOCUMENTS.
2. SHEET METAL DRAWINGS SHALL BE FULLY DIMENSIONED AND COORDINATED BASED ON FIELD VERIFIED BUILDING CLEARANCES AND ARCHITECTURAL CEILING LAYOUTS. INDICATE STRUCTURAL, LIGHTING, DUCTWORK AND PIPING AT ALL CRITICAL LOCATIONS.
3. CONTRACTOR SHALL REVIEW AND INDICATE HIS APPROVAL OF EACH SHOP DRAWING PRIOR TO SUBMITTAL FOR REVIEW. SHOP DRAWINGS WILL NOT BE REVIEWED BY THE ENGINEER UNLESS THE CONTRACTOR'S APPROVAL IS NOTED. DO NOT START WORK OR FABRICATION UNTIL SHOP DRAWINGS HAVE BEEN REVIEWED BY THE ENGINEER AND RETURNED TO THE CONTRACTOR.
4. SUBMITTALS WILL BE REVIEWED ONLY FOR GENERAL COMPLIANCE WITH THE CONTRACT DOCUMENTS AND NOT FOR DIMENSIONS OR QUANTITIES. THE SUBMITTAL REVIEW SHALL NOT RELIEVE THE CONTRACTOR OF RESPONSIBILITY FOR PURCHASE OF ANY ITEM IN FULL COMPLIANCE WITH THE CONTRACT DOCUMENTS OR ITS COMPLETE AND PROPER INSTALLATION.
5. WHERE SUBMITTALS VARY FROM THE CONTRACT REQUIREMENTS, THE CONTRACTOR SHALL CLEARLY INDICATE ON SUBMITTAL OR ACCOMPANYING DOCUMENTS THE NATURE AND REASON FOR THE VARIATIONS.
6. EACH MANUFACTURER OR HIS REPRESENTATIVE MUST CHECK THE APPLICATION OF HIS EQUIPMENT AND CERTIFY AT TIME OF SHOP DRAWING SUBMITTAL THAT THE EQUIPMENT SPECIFIED HAS BEEN PROPERLY APPLIED AND CAN BE INSTALLED, SERVICED AND MAINTAINED WHERE INDICATED ON THE DRAWINGS. ADVISE ENGINEER IN WRITING WITH SUBMITTAL DRAWINGS OF ANY POTENTIAL PROBLEMS. THE MANUFACTURER SHALL BE RESPONSIBLE FOR ANY CHANGES THAT MIGHT BE NECESSARY BECAUSE OF PHYSICAL CHARACTERISTICS OF EQUIPMENT THAT HAVE NOT BEEN CALLED TO THE ENGINEER'S ATTENTION AT THE TIME OF SUBMITTAL.

H. RECORD DRAWINGS

1. EACH CONTRACTOR OR SUBCONTRACTOR SHALL KEEP ONE (1) COMPLETE SET OF THE CONTRACT DRAWINGS ON THE JOB SITE ON WHICH HE SHALL REGULARLY RECORD ANY DEVIATIONS OR CHANGES FROM SUCH CONTRACT DRAWINGS MADE DURING CONSTRUCTION.
 2. THESE DRAWINGS SHALL RECORD THE INSTALLED LOCATION OF ALL CONCEALED EQUIPMENT, PIPING, ELECTRIC SERVICE, SEWERS, WASTES, VENTS, DUCTS, CONDUIT, ETC. BY MEASURED DIMENSIONS TO EACH SUCH ITEM FROM COLUMN CENTERLINES OR READILY IDENTIFIABLE AND ACCESSIBLE WALLS OR CORNERS OF THE BUILDING. PLANS ALSO SHALL SHOW INVERT ELEVATION OF SEWERS AND TOP ELEVATION OF ALL OTHER BELOW-GRADE LINES.
 3. RECORD DRAWINGS SHALL BE KEPT CLEAN AND UNDAMAGED AND SHALL NOT BE USED FOR ANY PURPOSE OTHER THAN RECORDING DEVIATIONS FROM WORKING DRAWINGS AND EXACT LOCATIONS OF CONCEALED WORK.
 4. AFTER THE PROJECT IS COMPLETED, THESE DRAWINGS SHALL BE DELIVERED TO THE ARCHITECT IN GOOD CONDITION, AS A PERMANENT RECORD OF THE INSTALLATION AS ACTUALLY CONSTRUCTED.
- BASIC MATERIALS AND METHODS**
- A. GENERAL
1. PROVIDE ALL MATERIALS, LABOR, EQUIPMENT, AND ACCESSORIES REQUIRED TO FURNISH AND INSTALL THE MECHANICAL ITEMS IDENTIFIED IN THIS SECTION.
 2. THIS SECTION INCLUDES BASIC MECHANICAL MATERIALS AND METHODS TO COMPLEMENT OTHER SECTIONS IN THIS SPECIFICATION AND REQUIREMENTS INDICATED ON THE MECHANICAL DRAWINGS.
- B. SUPPORTS AND HANGERS
1. INSTALL HANGERS, SUPPORTS, CLAMPS AND ATTACHMENTS AS REQUIRED TO PROPERLY SUPPORT PIPING FROM BUILDING STRUCTURE. THE INSTALLATION SHALL COMPLY WITH THE REQUIREMENTS OF MSS SP-69 AND SP-89.
 - C. ESCUTCHEONS
 1. FIT ALL PIPE PASSING THROUGH WALLS, FLOORS OR CEILINGS IN FINISHED ROOMS WITH STEEL OR BRASS ESCUTCHEONS. WHERE SURFACE IS TO RECEIVE A PAINT FINISH, MAKE ESCUTCHEONS PRIME PAINTED; OTHERWISE MAKE ESCUTCHEONS NICKEL OR CHROME PLATED. WHERE PIPING IS INSULATED, FIT ESCUTCHEONS OUTSIDE INSULATION.
- D. IDENTIFYING DEVICES AND LABELS
1. PROVIDE METAL EQUIPMENT NAMEPLATES PERMANENTLY FASTENED TO EQUIPMENT WITH OPERATIONAL DATA ENGRAVED OR STAMPED.
 2. IDENTIFY ALL PIPES AND VALVES IN EXPOSED AREAS, AND ACCESSIBLE CEILINGS AND SHAFTS.
 3. COLOR CODE IDENTIFICATION BANDS OR MARKER BACKGROUNDS TO IDENTIFY CONTENTS OF PIPE AND DIRECTION OF FLOW LOCATED NEAR EACH VALVE AND FITTING, ON BOTH SIDES OF PIPE PASSING THROUGH WALLS AND ON LONG RUNS AT NOT OVER 20 FOOT INTERVALS.

INSULATION

A. GENERAL

1. FURNISH ALL MATERIAL, LABOR AND EQUIPMENT AS REQUIRED TO INSTALL COMPLETE PLUMBING AND HVAC INSULATION AS INDICATED ON MECHANICAL DRAWINGS AND IN THESE SPECIFICATIONS.
 2. INSTALL IN FULL ACCORDANCE WITH MANUFACTURER'S RECOMMENDATIONS.
- B. INSULATION (AS MANUFACTURED BY OWENS CORNING, KNAUF, SCHULLER OR CERTAINTeed)
1. INSULATE ALL ABOVE GRADE HOT AND COLD WATER PIPING WITH MOLDED FIBER GLASS HAVING AN ALL SERVICE JACKET.
 - C. INSULATION THICKNESS SCHEDULE: 1) LESS THAN 2" DIAMETER PIPE: 1" THICK; 2) 2"-8" DIAMETER PIPE: 1-1/2" THICK.
 2. INCLUDE INSULATION OF FITTINGS AND VALVES. KEEP VAPOR BARRIERS INTACT. CONFORM TO MANUFACTURER'S RECOMMENDATIONS.
 - D. HVAC INSULATION (AS MANUFACTURED BY OWENS CORNING, KNAUF, SCHULLER, OR CERTAINTeed)
 1. INSULATE ALL NON-LINED SUPPLY, RETURN, AND EXHAUST DUCTS WITH 1-1/2" THICK FOIL FACED REINFORCED KRAFT JACKET, FIBERGLASS DUCT WRAP FULLY SECURED TO DUCT, R-6 MINIMUM. LAP AND TAPE SEAMS AND SECURE TIGHTLY TO THE DUCTS WITH WIRE OR STICK PINS. ALL INSULATION TO BE APPLIED IN FULL ACCORDANCE WITH THE MANUFACTURER'S RECOMMENDATIONS AND COMPLY WITH 25/50 FLAME AND SMOKE HAZARD RATINGS PER ASTM E-84, NFPA 255 AND UL 723.

PIPING AND VALVES

A. GENERAL

1. FURNISH ALL MATERIAL, LABOR, EQUIPMENT, AND ACCESSORIES AS REQUIRED TO INSTALL COMPLETE PLUMBING AND HVAC PIPING SYSTEMS AS INDICATED ON MECHANICAL DRAWINGS AND IN THESE SPECIFICATIONS.
 2. INSTALL IN FULL ACCORDANCE WITH LOCAL CODE REQUIREMENTS, OTHER SPECIFICATION SECTION REQUIREMENTS AND MANUFACTURER'S RECOMMENDATIONS.
- B. CONNECTIONS TO EQUIPMENT FURNISHED BY OTHERS
1. PROVIDE VALVED WATER CONNECTION FOR EQUIPMENT FURNISHED BY OTHER CONTRACTORS OR OWNER.
 2. INCLUDE ACCESSORIES REQUIRED BY CODE, DRAWING OR MANUFACTURER'S INSTRUCTIONS.
 - C. DOMESTIC WATER PIPING SHALL BE AS FOLLOWS:
 1. ABOVE GRADE - CPVC PIPING MANUFACTURED FROM CPVC COMPOUNDS WITH A CELL CLASS OF 24448 PER ASTM D1784.
 - a. FLOWGUARD GOLD CPVC CTS, MANUFACTURED TO SDR 11, CONFORMING TO ASTM D2846.
 - b. FITTINGS: SOCKET-TYPE, SOLVENT WELD JOINTS SHALL BE MADE USING CPVC CEMENT CONFORMING TO ASTM F493.
 - c. PIPE AND FITTINGS SHALL MEET OR EXCEED FLAME AND SMOKE REQUIREMENTS OF UL 723/ASTM E84.
 2. GRADE - TYPE "K" SOFT COPPER WITHOUT SPLICES OR FITTINGS.
 3. FLUSH, VENT AND SANITIZE ALL WATER PIPING WITH CHLORINE AS REQUIRED PER AWWA, LOCAL BUILDING DEPARTMENT AND HEALTH DEPARTMENT CODES.
 - D. CONDENSATE DRAIN PIPING
 1. INSTALL TYPE "M" COPPER CONDENSATE PIPING AT A MINIMUM 1% DOWNWARD SLOPE IN THE DIRECTION OF FLOW UNLESS OTHERWISE INDICATED. SCHEDULE 40 PVC PIPING SHALL BE AN ACCEPTABLE ALTERNATE.

PLUMBING FIXTURES AND EQUIPMENT

A. GENERAL

1. FURNISH ALL FIXTURES AND EQUIPMENT INDICATED AND SCHEDULED ON DRAWINGS, COMPLETE WITH ALL ACCESSORIES, CONTROLS, ETC., AS REQUIRED.
 2. INSTALL IN FULL ACCORDANCE WITH MANUFACTURER'S RECOMMENDATIONS AND PLACE IN SATISFACTORY OPERATION.
- B. TRAP PRIMER VALVES (AS MANUFACTURED BY PRECISION PLUMBING PRODUCTS, SIOUX CHIEF OR ZURN)
- HEAT GENERATION EQUIPMENT**
- A. GENERAL
1. FURNISH ALL MATERIAL, LABOR, EQUIPMENT AND ACCESSORIES AS REQUIRED TO INSTALL EQUIPMENT AS INDICATED ON MECHANICAL DRAWINGS.
 2. INSTALL IN FULL ACCORDANCE WITH LOCAL CODE REQUIREMENTS, OTHER SPECIFICATION SECTION REQUIREMENTS, AND MANUFACTURER RECOMMENDATIONS.
 - B. SEE EQUIPMENT SCHEDULES FOR ELECTRIC WATER HEATERS.

HVAC SYSTEMS AND EQUIPMENT

A. GENERAL

1. FURNISH ALL EQUIPMENT, MATERIALS, LABOR, TOOLS, ETC., FOR THE COMPLETE HVAC SYSTEM.
 2. INSTALL IN FULL ACCORDANCE WITH MANUFACTURER'S RECOMMENDATIONS AND PLACE IN SATISFACTORY OPERATION.
 3. CONTRACTORS BIDDING THIS PROJECT SHALL VISIT THE SITE AND FAMILIARIZE THEMSELVES WITH ALL CONDITIONS AFFECTING THEIR WORK. SUBMISSION OF A BID ON THIS PROJECT SHALL BE CONSTRUED AS HAVING SUCH KNOWLEDGE.
 4. VERIFY EXACT CONDITIONS IN FIELD AND COORDINATE WITH THESE DRAWINGS AND OTHER TRADES BEFORE BEGINNING NEW WORK.
 5. DETERMINE EXACT LOCATIONS FOR ALL EQUIPMENT, PIPING, CONDUITS AND DUCTWORK IN FIELD.
 6. COORDINATE WORK OF THIS CONTRACT WITH OTHER TRADES. CONFLICTS SHALL IMMEDIATELY BE BROUGHT TO THE ATTENTION OF THE ARCHITECT. ARCHITECT'S RESOLUTION TO CONFLICTS SHALL BE FINAL.
 7. DISCREPANCIES BETWEEN WHAT IS SHOWN ON DRAWINGS OR SPECIFIED AND THE ACTUAL CONDITIONS IN THE FIELD SHALL IMMEDIATELY BE BROUGHT TO THE ATTENTION OF THE ARCHITECT BEFORE PROCEEDING.
 8. BUILDING AND SURFACES DAMAGED DURING INSTALLATION SHALL BE REPAIRED, REPLACED, AND/OR RESTORED TO ORIGINAL CONDITION AFTER COMPLETION OF WORK AND BEFORE ACCEPTANCE BY OWNER.
- B. EQUIPMENT
1. MECHANICAL CONTRACTOR TO FURNISH ALL HVAC EQUIPMENT INDICATED AND/OR SCHEDULED ON THE DRAWINGS COMPLETE WITH BASES, ISOLATORS, SUPPORTS AND OTHER REQUIRED ACCESSORIES.
 2. INSTALL COMPLETE SYSTEMS AND PLACE IN PROPER OPERATION PER MANUFACTURER'S RECOMMENDATIONS. LUBRICATE AND ADJUST AS REQUIRED. FURNISH AND INSTALL CLEAN SET OF FILTERS PRIOR TO BALANCING.
 3. EQUIPMENT TO BE MAKE AND MODEL AS SCHEDULED UNLESS ALTERNATE EQUIPMENT OF EQUIVALENT QUALITY AND PERFORMANCE IS SUBMITTED AS A SUBSTITUTION PRIOR TO BIDDING. ALL SUBSTITUTIONS ARE SUBJECT TO ACCEPTANCE WITHOUT QUALIFICATION BY OWNER, ENGINEER AND ARCHITECT.
 4. FAN COIL UNITS SHALL MEET OR EXCEED THE MINIMUM EFFICIENCY REQUIREMENTS FOR AIR COOLED SPLIT SYSTEM HEAT PUMPS AS PER ASHRAE STANDARD 90.1.
 - C. FAN COIL UNIT CONDENSATE DRAINS
 1. PROVIDE EACH FAN COIL UNIT WITH A TRAPPED CONDENSATE DRAIN AS DETAILED ON THE DRAWING.
 - D. SEE EQUIPMENT SCHEDULES ON MECHANICAL DRAWINGS

AIR DISTRIBUTION SYSTEMS

A. GENERAL

1. FURNISH ALL MATERIALS, LABOR, EQUIPMENT AND ACCESSORIES REQUIRED TO INSTALL COMPLETE AIR DISTRIBUTION SYSTEMS.
 2. CONTRACTORS BIDDING THIS PROJECT SHALL VISIT THE SITE AND FAMILIARIZE THEMSELVES WITH ALL CONDITIONS AFFECTING THEIR WORK. SUBMISSION OF A BID ON THIS PROJECT SHALL BE CONSTRUED AS HAVING SUCH KNOWLEDGE.
 3. VERIFY EXACT CONDITIONS IN FIELD AND COORDINATE WITH THE DRAWINGS AND OTHER TRADES BEFORE BEGINNING NEW WORK.
 4. DETERMINE EXACT LOCATIONS FOR ALL NEW DUCTWORK AND ACCESSORIES IN THE FIELD.
 5. COORDINATE WORK OF THIS CONTRACT WITH OTHER TRADES.
 6. ANY DISCREPANCIES BETWEEN WHAT IS SHOWN ON DRAWINGS OR SPECIFIED AND THE ACTUAL CONDITIONS IN THE FIELD SHALL IMMEDIATELY BE BROUGHT TO THE ATTENTION OF THE ARCHITECT BEFORE PROCEEDING.
 7. BUILDING AND SURFACES DAMAGED DURING INSTALLATION BUILDING AND SURFACES DAMAGED DURING INSTALLATION SHALL BE REPAIRED, REPLACED, AND/OR RESTORED TO ORIGINAL CONDITION AFTER COMPLETION OF WORK AND BEFORE ACCEPTANCE BY OWNER.
- B. DUCTWORK
1. FABRICATE AND ERECT ALL DUCTWORK TO SMACNA STANDARDS FOR GALVANIZED STEEL. COMPLY WITH NFPA 90A REQUIREMENTS.
 2. DUCTWORK SHALL BE SMACNA LOW PRESSURE CONSTRUCTION 2" STATIC PRESSURE RATING WITH SEAL CLASS B SEAMS AND JOINTS.
 3. INCLUDE ALL ACOUSTIC, AIRFOIL SHAPED PERFORATED ALUMINUM TURNING VANES, MANUAL DAMPERS, FLEXIBLE CONNECTORS, GRILLES AND DIFFUSERS, ACOUSTIC LINING, AND OTHER SHEET METAL ACCESSORIES FOR THE PROJECT.
 4. EXHAUST DUCT OUTLETS SHALL BE INSTALLED A MINIMUM OF 10 FEET FROM ALL OUTSIDE AIR INTAKES.

HVAC INSTRUMENTATION AND CONTROLS

A. GENERAL

1. FURNISH AND INSTALL COMPLETE TEMPERATURE CONTROL FOR ALL HVAC SYSTEMS.
2. PROVIDE NEW CONTROL DEVICES INCLUDING THERMOSTATS, STAGING RELAYS AND OTHER RELATED DEVICES FOR A COMPLETE OPERATIONAL SYSTEM PER THE OPERATING SEQUENCE AND INDUSTRY STANDARDS.
3. MOUNT ALL CONTROLS FURNISHED AS ACCESSORIES TO EQUIPMENT AND PROVIDE ALL CONTROL WIRING REQUIRED FOR PROPER OPERATION. ALL WIRING SHALL BE IN CONDUIT PER NEC AND LOCAL CODE REQUIREMENTS.
- B. SEQUENCE OF OPERATION
1. HEAT PUMP
- a. WALL MOUNTED THERMOSTAT FURNISHED AS ACCESSORY TO UNIT SHALL SEQUENCE HEATING AND COOLING. PROVIDE WITH SUB-BASE TO MANUALLY SELECT HEATING, COOLING, FAN ON-OFF, AUTO OPERATION.
- b. A LOW TEMPERATURE THERMOSTAT WILL DE-ENERGIZE THE SUPPLY FAN AND CLOSE THE OUTSIDE AIR DAMPER IF THE MIXED AIR TEMPERATURE IS SENSED AT 40 DEGREES F OR COLDER.
2. TOILET ROOM EXHAUST FAN
- a. Shall operate continuously.

TESTING, ADJUSTING AND BALANCING

A. GENERAL

1. AFTER INSTALLATION, CHECK ALL EQUIPMENT AND PERFORM START UP IN ACCORDANCE WITH THE MANUFACTURER'S INSTRUCTIONS.
2. ALL PIPING SHALL BE TESTED AND FREE OF LEAKS AS REQUIRED BY THE LOCAL AUTHORITY HAVING JURISDICTION.
3. WORK THAT IS SCHEDULED TO BE CONCEALED OR INSULATED SHALL REMAIN UNCOVERED UNTIL REQUIRED TESTS HAVE BEEN COMPLETED. IF THE CONSTRUCTION SCHEDULE REQUIRES, ARRANGE FOR TESTS ON SECTIONS OF THE SYSTEM AT A TIME.
4. BALANCE ALL SYSTEMS, CALIBRATE CONTROLS, CHECK FOR PROPER OPERATION AND SEQUENCE UNDER ALL CONDITIONS AND MAKE ALL NECESSARY ADJUSTMENTS.
5. INSTRUCT OWNER IN OPERATION OF SYSTEMS AND OPERATING AND MAINTENANCE MANUAL FOR ALL EQUIPMENT AND SYSTEMS.
6. AIR AND WATER BALANCE REPORT FROM INDEPENDENT AABC OR NEBB CERTIFIED SUBCONTRACTOR FOR ALL AIR AND WATER SYSTEMS PER AABC OR NEBB STANDARDS.
- B. BALANCING, START UP AND INSTRUCTIONS
1. AFTER EQUIPMENT IS PLACED IN OPERATION, SYSTEMS SHALL BE BALANCED TO WITHIN 10% OF DESIGN FLOW WITH REPORT SUBMITTED TO OWNER. BALANCING SHALL BE PERFORMED BY AN INDEPENDENT AABC OR NEBB CERTIFIED CONTRACTOR.
2. BALANCE THE AIR SYSTEMS PRIOR TO BALANCING HYDRONIC AND REFRIGERANT SYSTEMS.
3. TEST, ADJUST AND BALANCE COOLING SYSTEMS DURING SUMMER SEASON AND HEATING SYSTEMS DURING WINTER SEASON. BALANCE SYSTEMS WHEN THE OUTSIDE AIR CONDITIONS ARE WITHIN 5 DEGREES F WET BULB TEMPERATURE OF THE MAXIMUM SUMMER DESIGN CONDITION AND WITHIN 10 DEGREES F DRY BULB TEMPERATURE OF THE MINIMUM WINTER DESIGN CONDITION.
4. START UP AND PLACE ALL SYSTEMS IN OPERATION AND TAG ALL SWITCHES AND CONTROLS WITH PERMANENT LABELS INSTRUCT OWNER ON PROPER OPERATION AND PREVENTATIVE MAINTENANCE OF SYSTEM.
3. DOCUMENT FUNCTIONS TO BE TESTED, INCLUDING CALIBRATIONS AND ECONOMIZER CONTROLS.
4. STATE CONDITIONS UNDER WHICH TEST WILL BE PERFORMED
- a. DESIGN TESTING SHALL AFFIRM WINTER AND SUMMER DESIGN CONDITIONS.



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SEAL



THIS DRAWING SHALL NOT BE USED FOR CONSTRUCTION PURPOSES UNTIL THE SEAL AND SIGNATURE OF THE RESPONSIBLE REGISTRANT APPEARS ON THE DRAWING, AND PROPER PERMIT FORMS AND RELATED FEES ARE TRANSMITTED BY THE OWNER, OWNER'S AGENT OR CONTRACTOR TO THE AUTHORITY HAVING JURISDICTION.

CONSULTANT

CLIENT

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PROJECT

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REVISIONS

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PERMIT

TITLE

MECHANICAL
SPECIFICATIONS

SHEET

M1